BRAC UNIVERSITY
Inspiring Excellence

CSE422: Artificial intelligence

Quiz 03 [Summer 2026]

Time: **20 minutes**, Full Marks: **15**

A

Name:		ID:		Section:	

1. A bank uses two independent algorithms to flag potentially fraudulent credit card transactions:
   - Velocity Check Algorithm (A): flags transactions based on unusual spending speed
   - Geolocation Mismatch Algorithm (B): flags transactions based on location inconsistencies

Historical records show that 3% of all transactions are fraudulent.

If a transaction is fraudulent (F):	If a transaction is not fraudulent (¬F):
• P(A flags | F) = 0.85 • P(B flags | F) = 0.70	• P(A flags | ¬F) = 0.05 • P(B flags | ¬F) = 0.08

Assume that, given whether a transaction is fraudulent or not, the two algorithms operate independently. A transaction is selected at random.

(a) Find the probability that Algorithm A flags the transaction, but Algorithm B does not, given that the transaction is fraudulent. **[3]**

(b) Suppose a selected transaction was flagged by Algorithm A but not flagged by Algorithm B. Find the probability that the transaction is actually fraudulent. **[5]**

(a) $P(A, \neg B \mid F) = P(A \mid F) \cdot P(\neg B \mid F)$ (as A and B are independent given F)

$= 0.85 \times (1 - 0.7)$

$= 0.255$

(b) $P(F \mid A, \neg B) = \dfrac{P(A, \neg B \mid F) \cdot P(F)}{P(A, \neg B)}$ (using Bayes)

$= \dfrac{P(A, \neg B \mid F) \cdot P(F)}{P(A, \neg B \mid F) \cdot P(F) + P(A, \neg B \mid \neg F) \cdot P(\neg F)}$

$= \dfrac{0.255 \times 0.03}{0.255 \times 0.03 + 0.046 \times (1 - 0.03)}$

$= 0.1464$

$P(A, \neg B \mid \neg F)$

$= P(A \mid \neg F) \cdot P(\neg B \mid \neg F)$

$= 0.05 \times (1 - 0.08)$

$= 0.046$

2. Consider the following historical data on what types of cat to buy from a store.

Colour	Age	Breed	Buy?
Orange	4	Shorthair	Buy
Black	3	Shorthair	Buy
White	1	Persian	Buy
Black	1	Persian	Not buy
Black	5	Shorthair	Not buy
Orange	2	Shorthair	Buy
Orange	3	Persian	Buy
Black	2	Shorthair	Not buy
White	1	Persian	Buy
Orange	3	Shorthair	Not buy

Determine whether you should buy an **orange Persian cat of age <=2** using **Naive-Bayes** classifier. **[7]**

$$P(buy \mid orange, persian, age \le 2) = \frac{P(orange, persian, age \le 2 \mid buy) \cdot P(buy)}{P(orange, persian, age \le 2)}$$

$$= P(orange \mid buy) \cdot P(persian \mid buy) \cdot \frac{P(age \le 2 \mid buy)}{P(orange, persian, age \le 2)} \cdot P(buy)$$

$$= \frac{\frac{3}{6} \cdot \frac{3}{6} \cdot \frac{3}{6}}{a} \cdot \frac{6}{10} = \frac{3/40}{a} = \frac{0.075}{a}$$

$$P(\neg buy \mid orange, persian, age \le 2) = \frac{P(orange \mid \neg buy) \cdot P(persian \mid \neg buy) \cdot P(age \le 2 \mid \neg buy)}{a} \cdot P(\neg buy)$$

$$= \frac{\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{2}{4}}{a} \cdot \frac{4}{10} = \frac{1/80}{a} = \frac{0.0125}{a}$$

$\because \frac{0.075}{a} > \frac{0.0125}{a}$. should buy.

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B

Name:		ID:		Section:	

1. An airport security system uses two independent scanners to detect prohibited items in checked luggage:
   - X-Ray Density Scanner (X): flags bags based on unusual material density
   - Trace Chemical Sensor (S): flags bags based on chemical residue detection

Historical records show that **2%** of all checked bags contain a prohibited item.

If a bag contains a prohibited item (PI):	If a bag does not contain a prohibited item (¬PI):
• P(X flags | PI) = 0.92 • P(S flags | PI) = 0.65	• P(X flags | ¬PI) = 0.07 • P(S flags | ¬PI) = 0.04

Assume that, given whether a bag contains a prohibited item or not, the two scanners operate independently. A bag is selected at random.

(a) Find the probability that the X-Ray Density Scanner flags the bag, but the Trace Chemical Sensor does not, given that the bag contains a prohibited item. **[3]**

(b) Suppose a selected bag was flagged by the X-Ray Density Scanner but not by the Trace Chemical Sensor. Find the probability that the bag actually contains a prohibited item. **[5]**

(a) $P(X, \neg S \mid PI) = P(X \mid PI) \cdot P(\neg S \mid PI) = 0.92 \times (1 - 0.65) = 0.322$

(b) $P(PI \mid X, \neg S) = \dfrac{P(X, \neg S \mid PI) \cdot P(PI)}{P(X, \neg S)}$

$= \dfrac{P(X, \neg S \mid PI) \cdot P(PI)}{P(X, \neg S \mid PI) \cdot P(PI) + P(X, \neg S \mid \neg PI) \cdot P(\neg PI)}$

$= \dfrac{0.322 \times 0.02}{0.322 \times 0.02 + 0.0672 \times (1 - 0.02)}$

$= 0.0891$

$P(X, \neg S \mid \neg PI) = P(X \mid \neg PI) \cdot P(\neg S \mid \neg PI)$

$= 0.07 \times (1 - 0.04)$

$= 0.0672$

2. Consider the following historical data on what types of dog to buy from a store.

Colour	Age	Breed	Buy?
Brown	4	Shepherd	Buy
Black	3	Shepherd	Buy
White	1	Labrador	Buy
Black	1	Labrador	Not buy
Black	5	Shepherd	Not buy
Brown	2	Shepherd	Buy
Brown	3	Labrador	Buy
Black	2	Shepherd	Not buy
White	1	Labrador	Buy
Brown	3	Shepherd	Not buy

Determine whether you should buy a **Brown Labrador dog of age >2** using **Naive-Bayes** classifier. **[7]**

$$P(buy|brown.labrador,age>2) = \frac{P(brown|buy).P(labrador|buy).P(age>2|buy)}{P(brown.labrador,age>2)}.P(buy)$$

$$= \frac{\frac{3}{6}.\frac{3}{6}.\frac{3}{6}}{a}.\frac{6}{10} = \frac{3/40}{a} = \frac{0.075}{a}$$

$$P(\neg buy|brown.labrador,age>2) = \frac{P(brown|\neg buy).P(labrador|\neg buy).P(age>2|\neg buy)}{P(brown.labrador,age>2)}.P(\neg buy)$$

$$= \frac{\frac{1}{4}.\frac{1}{4}.\frac{2}{4}}{a}.\frac{4}{10} = \frac{1/80}{a} = \frac{0.0125}{a}$$

$\because \frac{0.075}{a} > \frac{0.0125}{a}$, should buy.